

ADDIS ABABA UNIVERSITY
DEPARTMENT OF STATISTICS
Probability & Statistics (Stat 2171)
Test

Time allowed: 1:30 hrs

Point: 30%

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Instructions: Answer all questions. Show all necessary calculations for workout questions. Use appropriate statistical formulas and clearly state your answers.

Part I. Multiple Choice Questions ($15 \times 1 = 15$ marks)

1. A researcher is interested in the travel time of AAU students to college. A group of 50 students is interviewed. Their mean travel time is 16.7 minutes. For this study the mean of 16.7 minutes is an example of a(n)
 - A. Parameter
 - B. Statistic
 - C. Population
 - D. Sample
2. Identification number belongs to which level of measurement scale
 - A. Nominal scale
 - B. Ordinal scale
 - C. Interval scale
 - D. Ratio scale
3. Which of the following is not true about presentation of data?
 - A. They have greater attraction
 - B. They facilitate comparison
 - C. They allow valid inference
 - D. They are easy to understand
4. When data are categorized as, for example, places of residence (rural, suburban, urban), the most appropriate measure of central tendency is the
 - A. Mean
 - B. Mode
 - C. Median
 - D. Range

The following data show the number of hours worked by 200 statistics students.

Number of Hours	Frequency
0 - 9	40
10 - 19	50
20 - 29	70
30 - 39	40

5. Refer to the above data. The class width for this distribution
 - A. is 9
 - B. is 10
 - C. is 11
 - D. varies from class to class
6. Refer to the above data. The number of students working 19 hours or less
 - A. is 40
 - B. is 50
 - C. is 90
 - D. cannot be determined without the original data
7. Refer to the above data. The units of measurement is
 - A. 0.5
 - B. 0.1
 - C. 10
 - D. 1
8. Which measure of central tendency is the most commonly used?
 - A. Mode
 - B. Median
 - C. Arithmetic mean
 - D. Quartile

9. A good measure of central tendency should ideally be:
 - A. Difficult to interpret
 - B. Based on all observations
 - C. Highly affected by extreme values
 - D. Undefined for some data sets
10. The arithmetic mean of 4, 6, 8 and 10 is:
 - A. 6
 - B. 7
 - C. 8
 - D. 9
11. Which measure is most appropriate when a distribution contains extreme observations?
 - A. Mean
 - B. Median
 - C. Geometric mean
 - D. Harmonic mean
12. The mode is the:
 - A. Middle observation
 - B. Most frequent observation
 - C. Arithmetic average
 - D. Smallest observation
13. For a data set with an even number of observations, the median is:
 - A. The first observation
 - B. The largest observation
 - C. The average of the two middle observations
 - D. The most frequent observation
14. Which mean is particularly appropriate for rates, speeds and ratios?
 - A. Arithmetic mean
 - B. Geometric mean
 - C. Harmonic mean
 - D. Median
15. For positive observations, the usual relationship among the three means is:
 - A. $AM < GM < HM$
 - B. $AM = GM = HM$ always
 - C. $AM > GM > HM$
 - D. $HM > AM > GM$
16. A frequency distribution organizes observations according to:
 - A. Only their means
 - B. Classes and frequencies
 - C. Only their medians
 - D. Regression coefficients
17. Which frequency distribution is suitable for a small set of numerical discrete data with a small range?
 - A. Ungrouped frequency distribution
 - B. Pie chart
 - C. Histogram only
 - D. Stacked bar chart
18. Which graph uses adjacent bars to represent frequencies of continuous data?
 - A. Pie chart
 - B. Simple bar chart
 - C. Histogram
 - D. Line-free table
19. In a histogram, the horizontal axis represents:
 - A. Class boundaries/classes
 - B. Cumulative frequency only
 - C. Percentages only
 - D. Mean values only
20. The first quartile Q_1 has approximately what proportion of observations below or equal to it?
 - A. 10%
 - B. 25%
 - C. 50%
 - D. 75%

Part II. Fill in the Blank ($3 \times 1 = 3$ marks)

21. The sum of the deviations of all observations from their arithmetic mean is _____.
22. A set of data divided into classes together with the number of observations in each class is called a _____.
23. The three quartiles divide an ordered data set into _____ equal parts.

Part III. Short Answer Questions ($2 + 3 = 5$ marks)

24. What is statistics?
25. Write at least three uses and three limitations of statistics.

Part IV. Workout / Problem-Solving Questions ($4 \times 3 = 12$ marks)

26. Calculate the arithmetic mean, median, and mode for the data: 13, 16, 18, 15, 20, 22, 15, 17, 19.
25. Interpret the three measures briefly. Also calculate the variance and standard deviation.
27. A student records the following scores: 12, 5, 22, 30, 7, 36, 14, 42, 15, 53, 25. Arrange the observations in ascending order and calculate the median, Q1, and Q3.
28. Age in year of 20 women who attended health education at Jimma Health center in 1986 is summarized in the table. What is the mean, median and mode age of these women.

Time (in seconds)	Number of students
23-26	3
27-30	4
31-34	3
35-38	5
39-42	5

29. A meteorologist is interested to study the consistency of temperatures in two cities during a given week collected the following data. For which city is the temperature more consistent, based on these data?

City 1: 25 24 23 26 17

City 2: 22 21 24 22 20